



Infectious Disease

Clostridioides difficile Infection and Risk of Acute Graft-versus-Host Disease among Allogeneic Hematopoietic Stem Cell Transplantation Recipients



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Clostridioides difficile infection (CDI) is a major cause of infectious diarrhea among allogeneic hematopoietic stem cell transplantation (allo-HSCT) recipients. The relationship between CDI and acute graft-versus-host disease (aGVHD) has been a topic of interest, as these 2 conditions may influence each other. We studied the temporal relationship of CDI to aGVHD in the first 100 days post-transplantation in a large cohort of allo-HSCT recipients. We performed a retrospective cohort study of adult patients undergoing their first allo-HSCT at our tertiary care medical center between January 1, 2010, and December 31, 2016. Patients were followed for CDI diagnosis, development of aGVHD, and vital status up to day +100 post-transplantation. Descriptive statistics and multivariate Cox models with CDI as a time-varying covariate and aGVHD and high-grade aGVHD as outcomes were used for data analyses. A total of 656 allo-HSCT recipients were included in the analysis. Of these, 419 (64%) developed aGVHD, and 111 (17%) were diagnosed with CDI within the first 100 days post-transplantation. CDI developed before the onset of aGVHD in 72 of the 84 allo-HSCT recipients (85%) with both CDI and aGVHD. Fidaxomicin was used in the treatment of 57 of the 111 CDI cases (50%), whereas vancomycin was used in 52 (47%). Most of the CDI cases (88%) were diagnosed in the peritransplantation period (between day -10 and day +10). The median time to the development of CDI and aGVHD was 3.5 days (range, -7 to 95 days) and 33 days (range, 9 to 98 days) post-transplantation, respectively. Using multivariate Cox model, the following predictors were significantly associated with the development of aGVHD: CDI (adjusted hazard ratio [aHR], 1.52; 95% confidence interval [CI], 1.17 to 1.97; $P = .0018$), transplantation from a matched related donor (MRD) compared with a matched unrelated donor (aHR, 0.68; 95% CI, 0.54 to 0.85; $P = .0003$), and myeloablative versus nonmyeloablative conditioning (aHR, 2.45; 95% CI, 1.80 to 3.34; $P < .0001$), adjusting for age, sex, race, underlying disease, cytomegalovirus CMV serostatus, transplant source, and receipt of antithymocyte globulin (ATG). There was no association between CDI and high-grade aGVHD after adjustment for age, underlying disease, transplant type, intensity of conditioning, and receipt of ATG (aHR, 1.59; 95% CI, 0.95 to 2.66; $P = .0755$). CDI after allo-HSCT is associated with increased risk of GVHD when no CDI prophylaxis was used. Further studies examining CDI preventive measures, including prophylaxis, as well as the preservation or reconstitution of the gastrointestinal microbiome in the setting of HSCT are warranted.

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INTRODUCTION

Clostridioides difficile infection (CDI) is the most common cause of infectious diarrhea among hospitalized patients [1]. Allogeneic hematopoietic stem cell transplantation (allo-HSCT) recipients appear to be one of the populations at highest risk for this infection, with rates of CDI exceeding 25% in some studies [2].

Some investigators have suggested that CDI is more common after aGVHD [3–5], but the association between CDI and the subsequent development of aGVHD is less clear. Understanding the relationship between CDI and aGVHD is of importance, as prevention of CDI could be one way to reduce rates of aGVHD. To address this question, we performed a retrospective cohort study to determine whether CDI is an independent risk factor for aGVHD among allo-HSCT recipients at our tertiary care center.

METHODS
Study Design

We conducted a retrospective cohort study of all adult patients who underwent allo-HSCT between January 1, 2010, and December 31, 2016, at

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our midwestern tertiary care center. We used our institutional HSCT database and the electronic medical records for case ascertainment. We excluded patients who had undergone their first allo-HSCT before the study period or experienced graft failure in the first 100 days post-transplantation. We also excluded patients who had CDI diagnosed between day -30 and day -10 from stem cell infusion to decrease the likelihood of including patients with *C. difficile* colonization and patients who might still be receiving oral vancomycin for treatment. Patients were followed for CDI diagnosis, development of aGVHD, and vital status up to day +100 post-transplantation. Patient and transplantation characteristics were abstracted manually from the electronic medical records. The study was approved by the Institutional Review Board at our institution.

Definitions

CDI was defined as a physician diagnosis of diarrhea plus a positive *C. difficile* test between day -10 and day +100 of allo-HSCT. During the study period, our institution used several *C. difficile* tests: (a) C. Diff Quik Chek -TechLab, Blacksburg VA; (b) BD GeneOhm Cdiff assay, BD Diagnostics, San Diego, CA; (c) Portrait Toxigenic *C. difficile* Assay, Great Basin, West Valley City, UT, stool toxin A/B (EIA) Tox A/B Quick Check from January 2010 to May 2010; *C. difficile* toxin B PCR (NAAT) BD GeneOhm C diff Assay, from June 2010 to March 2014, and *C. difficile* toxin B PCR (NAAT) Great Basin Portrait Analyzer from April 2014 to December 2016.

During the early part of the study period, vancomycin was the standard antimicrobial used to treat CDI. Fidaxomicin became the standard treatment starting in June 2013. Staging and grading of aGVHD were determined using the standard Minnesota Center for International Blood and Marrow Transplant Research grading system [6], using clinical and pathological information between day 0 and day +100. Grades I and II were considered low-grade aGVHD, whereas grades III and IV were considered high-grade aGVHD.

Statistical Analysis

Descriptive statistics were used for the characteristics of the total allo-HSCT recipients included in the analysis. Predictors of aGVHD, high-grade aGVHD, and gastrointestinal (GI) aGVHD were analyzed using univariate Cox models; predefined variables and those with $P \leq .1$ were included in multiple regression Cox models for aGVHD and high-grade aGVHD. CDI was defined as a time-varying covariate to censor episodes that occurred after a diagnosis of aGVHD. Predefined and variables significant at $P \leq .05$ were retained in the final models after excluding colinear predictors. All statistical analyses were conducted in SAS 9.4 (SAS Institute, Cary, NC).

RESULTS

Patient Characteristics

Between January 1, 2010, and December 31, 2016, 679 patients underwent allo-HSCT at our institution. Only those undergoing initial transplantation were included in this analysis. Of the 679 patients, 23 were excluded, including 8 patients who had undergone their first allo-HSCT before the study period, 5 who experienced graft failure within the first 100 days post-transplantation, 4 who were age <18 years, 2 who had CDI diagnosed between day -30 and day -10, and 4 who had incomplete data. A total of 656 allo-HSCT recipients were included in the analysis.

Patient demographic characteristics, underlying diseases, and transplantation characteristics are shown in Table 1. The mean patient age at the time of transplantation was 50 years (range, 18 to 75 years). The 2 most common indications for transplantation were acute myelogenous leukemia, in 279 patients (43%), and myelodysplastic syndrome, in 110 patients (17%). Peripheral stem cells served as the graft source in the majority of patients (503; 77%). Matched unrelated donor (MUD) transplants were performed in 311 patients (48%), and matched related donor (MRD) transplants were performed in 219 (33%). Cord blood and haploidentical transplants accounted for only small percentages of the transplants, 67 (10%) and 59 (9%), respectively. The conditioning regimen was reduced intensity in 262 patients (40%), myeloablative in 252 (38%), and nonmyeloablative in 142 (22%). During the study period, the most commonly used GVHD prophylaxis regimen was tacrolimus and methotrexate, followed by tacrolimus and mycophenolate mofetil. Death occurred in 62 patients (9%)

Table 1

Baseline Characteristics of the Allo-HSCT Recipients (n = 656)

Characteristic	Value
Age, yr, median (range)	50 (18–75)
Sex, n (%)	
Male	400 (61)
Female	256 (39)
Race, n (%)	
White	565 (86)
African American	41 (6)
Other	50 (8)
Underlying disease, n (%)	
AML	279 (43)
MDS	110 (17)
Lymphoma	100 (15)
ALL	70 (11)
Others	97 (14)
Transplant type, n (%)	
MUD	311 (48)
MRD	219 (33)
Cord blood	67 (10)
Haploidentical	59 (9)
Stem cell source, n (%)	
Peripheral blood	503 (77)
Bone marrow	86 (13)
Cord blood	67 (10)
Conditioning regimen intensity, n (%)	
Myeloablative	252 (38)
Reduced intensity	262 (40)
Nonmyeloablative	142 (22)
CMV serostatus, n (%)	
Donor ⁺ /recipient ⁺	220 (34)
Others	436 (66)

AML indicates acute myelogenous leukemia; MDS, myelodysplastic syndrome; ALL, acute lymphoblastic leukemia.

before day +100, with the most common causes being transplantation-related complications (ie, aGVHD and infections) and disease progression or relapse.

CDI and aGVHD

aGVHD was diagnosed in 419 of 656 patients (64%) during the first 100 days post-transplantation. Diagnosis was based on clinical criteria, with pathological supportive criteria in some of the patients. The median time to aGVHD diagnosis was 33 days (range, 9 to 98 days). aGVHD was classified as low-grade (grade I/II) in 319 patients (76%), high-grade (III/IV) in 92 (22%), and undocumented grade in 8 (2%). Gastrointestinal (GI) aGVHD alone or in association with another organ system was diagnosed in the majority of patients (75%) with aGVHD. Biopsies specimens of GI, skin, or liver were obtained to evaluate for aGVHD in 373 patients (57%). GI biopsies were performed in 281 of the 656 patients (42%) and were supportive of GI aGVHD in 132 (47%), inconclusive in 25 (9%), and negative in 124 (44%).

CDI was diagnosed in 111 of the 656 (17%) allo-HSCT recipients. The median time to CDI onset was 3.5 days from transplantation (range, day -7 to day +95). Fidaxomicin was used to treat CDI in 57 patients (50%), and vancomycin alone was used in 53 (47%). Among the patients with CDI, 84 (76%) developed aGVHD, with CDI developing before the onset of aGVHD in 72 (86%) of those cases (Table 2). Among patients with CDI before aGVHD, the median time to diagnosis of CDI and aGVHD from

Table 2
CDI and aGVHD Characteristics in the Allo-HSCT Recipients

Characteristic	Value
Total CDI cases, n (%)	111 (100)
CDI timing, n (%)	
CDI before aGVHD	72 (64)
CDI after aGVHD	11 (10)
CDI with no aGVHD	27 (25)
CDI and aGVHD on same date	1 (1)
CDI treatment, n (%)	
Fidaxomicin	57 (51)
Vancomycin	52 (47)
Other	2 (2)
Type of aGVHD, n (%)	
Total aGVHD cases	419 (100)
Skin + GI	151 (36)
GI	130 (31)
Skin	94 (22)
Other	44 (11)
aGVHD grade, n (%)	
I	79 (19)
II	240 (57)
III	59 (14)
IV	33 (8)
Biopsy-proven aGVHD, n (%)	185 (44)
Death within 100 d, n (%)	62 (9)
Median time to aGVHD, d	33
Median time to CDI, d	3.5

day 0 of allo-HSCT were 1 day (range, -7 to +39 days) and 29 days (range, +9 to +96 days), respectively. In patients with CDI after aGVHD, the median time to aGVHD and CDI were 28 days (range, +20 to +86 days) and 55 days (range, +37 to +95 days) following allo-HSCT, respectively. High-grade (grade III/IV) aGVHD was noted in 18 of 72 patients (25%) who had CDI before aGVHD, compared with 68 of 545 patients (12%) without CDI.

Only 8 patients had more than one CDI episode from day -10 to +100 post-HSCT. Four of these patients experienced relapse within 8 weeks from the first CDI. Seven of the 8 patients developed aGVHD. All the patients survived to 100 days.

In univariate Cox models, the following predictors were associated with development of aGVHD: CDI (HR, 1.64; 95% CI, 1.28 to 2.11; $P = .0001$), MRD versus MUD (HR, 0.70; 95% CI, 0.56 to 0.87; $P = .0015$), haploidentical donor versus MUD (HR, 0.74; 95% CI, 0.52 to 1.05; $P = .0940$), and myeloablative versus nonmyeloablative conditioning (HR, 2.25; 95% CI, 1.72 to 2.94; $P < .0001$). Age, sex, race, underlying disease, CMV serostatus, and use of antithymocyte globulin (ATG) were not associated with higher risk of aGVHD (Table 3). In a multivariate Cox model, the following predictors remained significantly associated with aGVHD: CDI (adjusted hazard ratio [aHR], 1.52; 95% CI, 1.17 to 1.97; $P = .0018$), MRD compared with MUD (aHR = 0.68, 95% CI 0.54 – 0.85, $p = 0.0003$), myeloablative versus non-ablative conditioning (aHR = 2.45, 95% CI, 1.80 to 3.34; $P < .0001$) adjusting for age, sex, race, underlying disease, CMV serostatus, transplant source, and receipt of ATG (Table 3).

In additional univariate Cox models, CDI was associated with high-grade aGVHD (HR, 1.82; 95% CI, 1.11 to 2.99; $P = .0174$) but was no longer significant after adjustment for age, underlying disease, transplant type, and conditioning

regimen intensity (aHR, 1.59; 95% CI, 0.95 to 2.66; $P = .0755$) (Table 5).

On unadjusted analysis, CDI was also associated with GI aGVHD (HR, 1.76; 95% CI, 1.33 to 2.34; $P = .0174$). Furthermore, among the 72 patients with CDI before aGVHD, treatment with fidaxomicin was associated with a lower risk of GI aGVHD compared with treatment with vancomycin (unadjusted HR, 0.40; 95% CI, 0.24 to 0.69; $P = .0010$). There was no statistically significant association between treatment groups (fidaxomicin versus vancomycin) and the development of overall aGVHD.

DISCUSSION

The reason why allo-HSCT recipients have an increased risk for developing CDI compared with the general hospitalized population is not fully understood but is likely explained by dysregulation of the gut microbiota, mainly through cytotoxic chemotherapy, immunosuppression, and antimicrobial exposure [7]. In addition to an increased risk for CDI, alterations in the GI microbiome during the transplantation course also may influence the risk for other transplantation complications, such as bacteremia [8] and the development of aGVHD [9]. The development of aGVHD is a complex multistep process that involves damage to the GI mucosa by conditioning chemotherapy and/or radiotherapy regimens [10]. Damaged host tissues secrete proinflammatory cytokines that activate host antigen-presenting cells, leading to donor T cell activation, proliferation, and induction of cellular mediators, such as TNF- α and other cytokines, and resulting in further tissue damage [11]. GI dysbiosis also may trigger the activation of host antigen-presenting cells, leading to immune dysregulation, alterations in the integrity of the intestinal tight junctions, and the development of aGVHD [12,13].

With recognition of the close association of intestinal dysbiosis and immune and inflammatory conditions such as aGVHD, the relationship between CDI and aGVHD has been a topic of great interest, as either of the 2 problems can be expected to influence the other. Previous studies have addressed the incidence, risk factors, and outcomes of CDI after allo-HSCT [3,14–18]; however, data on the association between CDI and the subsequent development of aGVHD in allo-HSCT recipients is limited and contradictory, with some studies supporting this association [3,10,14] and others showing no significant correlation [4,16,18]. Notably, the largest study to date, published in 2018 by Bhutani et al. [19], retrospectively examined 310 allo-HSCT recipients and reported CDI to be statistically associated with a higher probability of subsequent GI aGVHD (HR, 1.92; 95% CI, 1.15 to 3.19; $P = .01$) (Table 4).

Our study of 656 allo-HSCT recipients is one of the largest retrospective studies designed to examine the effect of CDI on the development of aGVHD. Our findings support earlier observations that CDI is an independent predictor of aGVHD in the first 100 days after allo-HSCT. It is important to recognize that our observation was made when no CDI prophylaxis was used and does not imply direct causality between CDI and aGVHD. In our cohort, 419 (64%) of the allo-HSCT recipients developed aGVHD, with 92 (12%) experiencing high-grade aGVHD. Although there was a trend toward an association of CDI with high-grade aGVHD, this did not remain statistically significant when adjusting for other variables.

The GI tract was the most frequently involved organ, in 75% of aGVHD cases. In our population, roughly one-half of GI biopsies supported the diagnosis of GVHD with 10% being inconclusive. However, aGVHD is a clinical diagnosis and having negative or inconclusive biopsies cannot rule out the diagnosis. Although biomarkers may be an objective and reliable way to

Table 3

Predictors of aGVHD in Allo-HSCT Recipients, Univariate and Multivariate Cox Regression Analysis (n = 656)

Variable	Unadjusted HR (95% CI)	P Value	aHR (95% CI)	P Value
CDI				
No	Reference		Reference	
Yes	1.64 (1.28–2.11)	.0001	1.52 (1.17–1.97)	.0018
Age, yr	0.99 (0.98–1.00)	.1936	1.00 (0.99–1.01)	.6127
Sex				
Male	Reference		Reference	
Female	1.05 (0.87–1.28)	.6074	1.07 (0.88–1.31)	.4993
Race				
White	Reference		Reference	
Black	0.97 (0.65–0.45)	.8920	1.04 (0.69–1.56)	.8489
Other	0.90 (0.57–1.41)	.6425	0.89 (0.57–1.40)	.6223
Underlying disease				
Acute leukemia	Reference		Reference	
Lymphoma	1.26 (0.96–1.67)	.1014	1.03 (0.76–1.39)	.8532
Other	1.20 (0.97–1.50)	.1017	0.98 (0.78–1.24)	.8524
CMV status				
D+ or R+	Reference		Reference	
D-/R-	0.94 (0.77–1.16)	.5800	0.92 (0.75–1.13)	.4346
Transplant type				
MUD	Reference		Reference	
MRD	0.70 (0.56–0.87)	.0015	0.68 (0.54–0.85)	.0008
Haploidentical	0.74 (0.52–1.05)	.0940	0.94 (0.62–1.42)	.7645
Cord	0.83 (0.59–1.16)	.2754	0.63 (0.29–1.36)	.2369
Transplant source				
Peripheral blood	Reference		Reference	
Bone marrow	0.82 (0.61–1.10)	.1876	0.81 (0.58–1.13)	.2056
Cord blood	0.65 (0.31–1.38)	.2639	1.55 (0.68–3.54)	.2951
Regimen intensity				
Nonmyeloablative	Reference		Reference	
Myeloablative	2.25 (1.72–2.94)	<.0001	2.45 (1.80–3.34)	<.0001
Reduced intensity	1.17 (0.89–1.54)	.2574	1.26 (0.92–1.73)	.1454
ATG use				
No	Reference		Reference	
Yes	0.92 (0.68–1.26)	.6046	0.79 (0.569–1.10)	.1544

diagnose and predict outcomes of aGVHD [20,21], their clinical use in diagnosing aGVHD remains limited. Therefore, differentiating GI GVHD from other causes of diarrhea in this group of patients remains challenging and may lead to overestimation of GI aGVHD. This observation may have influenced the ability to demonstrate a clear association of CDI and GI aGVHD after adjusting for other variables. Nonetheless, it is notable that in our study, the median time to CDI was much shorter than the median time to aGVHD, making it unlikely that CDI and aGVHD were overlapping.

Our 17% rate of CDI after allo-HSCT is comparable to that in previous reports [1]. Similarly, our study confirms that the greatest risk for CDI is during the preengraftment period [5]. Approximately 88% of CDI cases were diagnosed between day -10 and day +10 from stem cell infusion in our cohort. The median time to CDI in our patients was significantly shorter than reported in the literature (12 to >30 days) [3,19,22]. Some of this variability is reflective of the duration of follow-up in these studies. Bhutani et al. [19] evaluated 310 allo-HSCT recipients with a median follow-up of 2.43 years. The median time to CDI was 12 days (range, -3 to 113 days) in the related donor group and 16 days (range, -3 to 209 days) in the unrelated donor group. In the present study, we limited follow-up to 100 days post-transplantation, and thus we did not account

for episodes of CDI that may have occurred beyond 100 days, as we were interested in examining the association of CDI with aGVHD. Even with this consideration, the median time to CDI in our study was only 3 days. The apparent high rate of CDI in this early post-transplantation period may be explained by potentially including a large number of allo-HSCT recipients colonized with *C difficile* who developed diarrhea due to other causes, such as chemotoxicity and prophylactic antibiotics. Our study did not examine the rate of *C difficile* colonization before or during the study period. We attempted to minimize this effect by excluding patients with CDI diagnosed within 30 days of allo-HSCT. Differentiating between *C difficile* colonization and CDI is difficult; however, it is also possible that *C difficile* colonization may have led to an early occurrence of CDI. *C difficile* colonization before allo-HSCT has been identified as a significant risk factor for the subsequent development of CDI. Bruminhent et al. [23] showed that in patients undergoing allo-HSCT, *C difficile* colonization on admission was a significant risk factor for the subsequent development of CDI (odds ratio, 68.5; 95% CI, 11.4 to 416.2). This finding appears to contradict the observation that CDI rates are decreased in the general population among asymptomatic *C difficile* carriers who develop an adequate immune response [24]. Whether this represents a unique feature of immune dysregulation

Table 4

Studies Evaluating the Association between GVHD and CDI

Authors	Year	Design	Participants	Purpose	Results
Chakrabarti et al. [14]	2000	Retrospective cohort	75 allo-HSCT recipients	To study the incidence, risk factors, and outcome of infection with <i>C difficile</i> in HSCT	Using univariate analysis, a strong association was found between CDI and GVHD. The strongest association in this study was with grade III and IV GVHD (5/10 versus 6/65 uninfected patients; $P = .004$; odds ratio, 9.8).
Dubberke et al. [15]	2010	Case-control study	37 CDI cases and 67 controls	To evaluate risk factors and outcomes associated with CDI in allo-SCT recipients	Patients with CDI were more likely to develop new-onset GVHD ($P < .001$), new-onset severe GVHD ($P < .001$), or new-onset gut GVHD ($P = .007$).
Alonso et al. [3]	2012	Retrospective cohort	999 adult HSCT recipients (allogeneic and autologous)	To study the epidemiology and outcomes of CDI in HSCT recipients	Using univariate analysis, patients who developed CDI were more likely to develop GI GVHD compared with patients who never developed CDI (log-rank test; $P = .001$).
Trifilio et al. [17]	2013	Retrospective cohort	822 HSCT recipients (autologous and allogeneic)	To study the epidemiology and outcomes of CDI in HSCT recipients	CDI patients were more likely to develop severe GVHD at day 60 and day 100 after HSCT.
Willems et al. [4]	2012	Retrospective cohort	407 allo-HSCT recipients	To evaluate the incidence, risk factors, and outcome of CDI after HSCT	Using univariate analysis, CDI was not found to be significantly associated with the risk of occurrence of a subsequent grade $\geq$ II aGVHD (HR, 1.24; 95% CI, 0.72–2.15; $P = .43$) or GI aGVHD (HR, 1.23; 95% CI, 0.67–2.28; $P = .51$).
Kinnebrew et al. [16]	2014	Randomized clinical trial	94 allo-SCT	To determine the outcome of intestinal colonization of toxigenic <i>C difficile</i>	No association was found between early or later post-transplantation CDI and GVHD.
Kamboj et al. [18]	2014	Retrospective cohort	793 allo-HSCT recipients (adult and pediatric)	To study the incidence, risk factors, and clinical outcomes, including GVHD associated with early CDI in allo-HSCT recipients	The cumulative incidence of GI GVHD in patients with and without early CDI was 6.8% and 8%, respectively ($P = .5$), and there was no significant association between early CDI and subsequent development of GI GVHD.
Bruminhent et al. [23]	2014	Retrospective cohort	150 HSCT recipients (autologous and allogeneic)	To determine the effect of <i>C difficile</i> colonization on hospital-acquired CDI	No significant differences in rates of GI or skin GVHD between <i>C difficile</i> colonization and non- <i>C difficile</i> colonization patients or patients with and without CDI.
Dubberke et al. [22]	2017	Prospective cohort	187 allo-HSCT recipients	To evaluate risk factors and outcomes of CDI in allo-HSCT recipients	GVHD of any kind was associated with significantly increased risk of CDI in the postengraftment population.
Bhutani et al. [19]	2019	Retrospective cohort	310 allo-HSCT recipients	To study the relationship between CDI and GI GVHD	Patients who developed CDI had a statistically significant higher probability of developing GI aGVHD (HR, 1.92; 95% CI, 1.15–3.19; $P = .01$).

among allo-HSCT recipients compared with non-transplanted patients is uncertain and warrants further investigation.

Limitations of this study are its retrospective design and difficulty in determining the pretransplantation history of *C difficile* colonization or infection and systemic antibiotic exposure. These factors make differentiating between CDI and *C difficile* colonization difficult and further limit the ability to draw inference of causality in the association found between aGVHD and previous CDI. This study is also limited by the inability to

monitor alterations in the intestinal microbiome throughout the transplantation period. GI microbiome dysregulation unrelated to infections such as CDI appears to influence allo-HSCT complications, including GVHD [25,26].

Longitudinal monitoring of the microbiota in allo-HSCT recipients has shown that loss of diversity occurs early during transplantation, when patients receive chemotherapy, radiation, and donor cells [27]. This reduced diversity also correlates with decreases in commensal bacteria, which have been shown

Table 5

Predictors of High-Grade aGVHD in allo-HSCT Recipients, Univariate and Multivariate Cox Regression Analysis

Variable	Unadjusted HR (95% CI)	P Value	aHR (95% CI)	P Value
CDI				
No	Reference		Reference	
Yes	1.824 (1.111–2.995)	.0174	1.593 (0.953–2.661)	.0755
Age, yr	0.987 (0.974–1.001)	.06	0.996 (0.980–1.011)	.5825
Sex				
Male	Reference			
Female	0.603 (0.383–0.949)	.0288		
Race				
White	Reference			
Black	1.684 (0.844–3.360)	.1394		
Other	0.904 (0.331–2.472)	.8447		
Underlying disease				
Acute leukemia	Reference		Reference	
Lymphoma	0.897 (0.487–1.650)	.7257	1.213 (0.634–2.323)	.5596
Other	1.004 (0.636–1.587)	.9856	1.179 (0.721–1.928)	.5119
CMV status				
D+ or R+	Reference			
D-/R-	1.077 (0.701–1.654)	.7351		
Transplant type				
MUD	Reference		Reference	
MRD	0.654 (0.402–1.064)	.0874	0.680 (0.413–1.121)	.1309
Haploidentical	0.4 (0.145–1.108)	.078	0.465 (0.159–1.360)	.1619
Cord blood	1.356 (0.750–2.454)	.3136	1.779 (0.912–3.468)	.0909
Transplant source				
Peripheral blood	Reference			
Bone marrow	0.754 (0.377–1.511)	.4266		
Cord blood	0.971 (0.238–3.962)	.9675		
Regimen intensity				
Nonmyeloablative	Reference		Reference	
Myeloablative	1.988 (1.113–3.551)	.0202	2.138 (1.125–4.065)	.0204
Reduced intensity	1.078 (0.578–2.011)	.8135	1.450 (0.727–2.894)	.2919
ATG use				
No	Reference		Reference	
Yes	1.391 (0.788–2.457)	.255	1.136 (0.612–2.108)	.6865

to contribute to normal intestinal hemostasis, including barrier function and immune tolerance [28]. Similarly, GVHD severity and microbiome damage also may be influenced by antibiotic exposure. Antibiotics with strong activity against obligate anaerobes correlates with increased disruption of microbiome, higher intestinal barrier damage and more severe GVHD [29,30]. Conversely, preservation of the microbiome is associated with less GVHD. Abundance of *Blautia* spp, a member of the Clostridium class of Firmicutes has been shown to be associated with better survival, less GVHD and relapse-related GVHD, and less need for GVHD-related treatment [31]. The mechanism of this association is not clear, but increased abundance of *Blautia* is associated with the recovery of mucosal-associated invariant T cells, an immune regulatory population [32].

The majority of participants in our study received antibiotics as either prophylaxis or treatment during their transplantation course. The antibiotic prophylaxis regimen remained standardized during the study period. We did not include B-lactam antibiotic exposure in our analysis, as previous reports have suggested that B-lactam antibiotics are commonly given in the peritransplantation period [19]; however, we acknowledge that some differences in antibiotic exposure may have occurred and influenced the development of aGVHD that we did not identify.

As a result of standardization of CDI treatment at our institution approximately 50% of CDI cases were treated with fidaxomicin. On unadjusted analysis, treatment with fidaxomicin was associated with lower risk of GI aGVHD (but not overall aGVHD) compared to treatment with vancomycin. If proven, this supports previous observations in a murine model that fidaxomicin exposure had less impact on the microbiota and was associated with faster microbial recovery and less *C. difficile* colonization, compared to vancomycin [33]. Further studies are warranted to evaluate this important observation.

Approaches to either restore or preserve the intestinal microbiome during allo-HSCT may prove useful in decreasing GVHD. Recently reported data suggest that fecal microbiota transplantation (FMT) is a safe and effective procedure in patients before or after allo-HSCT [34]. Ongoing evaluation of microbiome alteration as a preventive measure by using FMT capsules early after allo-HSCT is expected to provide some additional insight into the association of microbiome and outcomes such as aGVHD. Antibiotic inactivation in the GI tract through the use of orally administered beta-lactamase represents a strategy to protect colonic microbiota and reduce antibiotic resistance. Ribaxamase, an orally administered beta-lactamase, has been shown to degrade ceftriaxone in the GI tract and preserve the gut microbiome [35]. Whether this

would be effective in preserving intestinal microbiome during the period of HSCT remains to be determined.

The diagnosis of CDI has many inherent limitations, particularly among a group of allo-HSCT recipients with a high likelihood of CDI colonization and diarrhea from various causes. During most of our study period, *C difficile* toxin B NAAT was used for diagnosis. Despite the high sensitivity of NAAT tests for CDI, they do not detect actual toxin production and can detect asymptomatic *C difficile* colonization. Using a 2-step approach to diagnosing CDI starting with a glutamate dehydrogenase toxin A/B enzyme immunoassay for toxin A or B and then moving to a PCR assay to detect toxigenic *C difficile* might be a helpful strategy to reduce detection and unnecessary treatment of *C difficile* colonization [36]. Whether *C difficile* colonization without infection has any direct role in the development of aGVHD or alters the inflammatory milieu of the gastrointestinal tract is uncertain and warrants further study.

Currently, there are no guideline recommendations for using CDI antimicrobial prophylaxis in allo-HSCT recipients. Whether CDI prophylaxis can prevent aGVHD is uncertain. In a study of 145 allo-HSCT recipients, primary prophylaxis with oral vancomycin was highly effective in preventing CDI with no effect on the incidence of aGVHD [37]. A large randomized study of fidaxomicin prophylaxis in HSCT recipients was not associated with a reduction in a composite outcome of CDI or aGVHD [38]. CDI prophylaxis was not used at our institution during the study period. In our study, during the 30 days before transplantation, only 2 of the 679 potential subjects were excluded, because they were completing treatment for recent CDI infections. The small number of excluded subjects is not likely to impact our findings.

In conclusion, allo-HSCT recipients with CDI had a 52% higher risk of subsequent aGVHD in our large retrospective cohort study after adjustment for age, underlying disease, transplant type, conditioning regimen intensity, and receipt of ATG. Measures to reduce CDI are needed. Although antibiotic prophylaxis against CDI is one potential approach and further studies are warranted, preserving or reconstituting the delicate balance of the GI microbiome in the setting of HSCT may be a more successful endeavor.

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